



Technical Service Bulletin: TSB220231

Released Date: 15-Apr-2026

Updated Turbocharger Service Parts for Turbine to Bearing Housing Bolted Joint

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK60 CM500
- QSK60 CM850 MCRS
- QSK60 CM2150 MCRS
- QSK60 CM2350 K116
- QSK60 CM2350 K121
- QSK60 CM2350 K135
- QSK60 CM2350 K136

Issue Summary

Symptom:

- Broken, loose, or missing turbine to bearing housing capscrews **only** on HE800FG turbochargers

Verification

Verify unit is equipped with HE800FG turbochargers utilizing either 16 mm [0.630 in] or 25 mm [0.984 in] length turbine to bearing housing capscrews by measuring the length of the capscrew. If the capscrew length is found to be 16 mm [0.630 in], verify if the turbocharger is equipped with a CSeal. See Figure 1 for identification of a turbocharger with a CSeal versus a turbocharger that does **not** contain a CSeal.

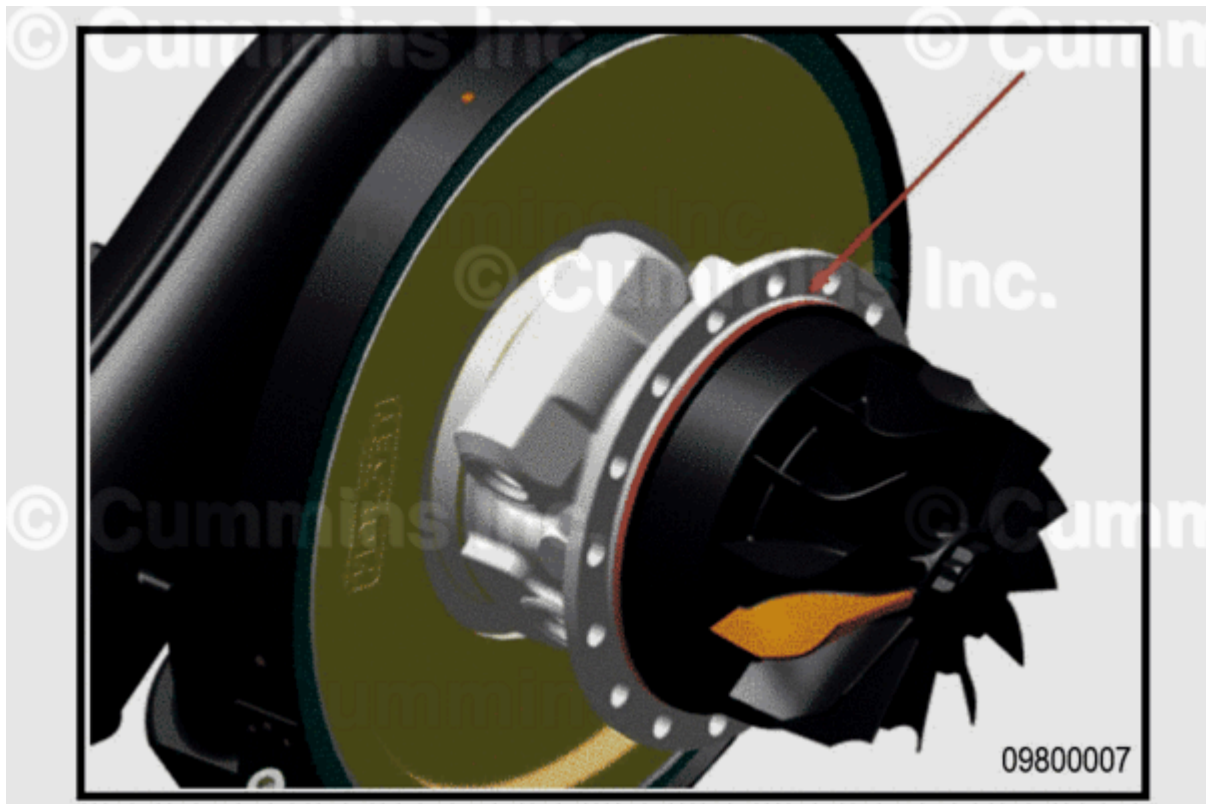


Figure 1, CSeal Turbocharger Identification

Resolution

Lockplates installed under B16 or Inconel capscrews improve torque retention of HE800FG turbochargers and reduce susceptibility to malfunction.

Service Instructions

- In the event of a HE800FG Turbocharger having broken, loose, or missing turbine to bearing housing capscrews, all eight (8) turbine to bearing capscrews **must** be replaced.
- Capscrew replacements will require the installation of lockplates underneath the capscrews. When the lockplate tabs are bent upwards after torquing the bolts, the lockplate tabs **must** seat on the flat sides of the capscrew head and **must not** be seated on the corner of the capscrew head. Refer to Figure 2. Additional torquing past the specified value up to a maximum of 60 degrees is allowed to align the next capscrew head flat with the lockplate tabs.
- If the turbocharger was originally fitted with 25 mm [0.984 in] capscrews, 25 mm [0.984 in] length capscrews **must** be used for replacement. The 25mm capscrew material will depend on the Service Model Name and will be either B16 or Inconel. Any turbocharger configuration utilizing 25 mm [0.984 in] capscrews **must not** be fitted with a shim between the flange of the bearing housing and turbine housing. Reference Table 1 for the appropriate bolted joint type for the engine based on the Service Model Name.
- If the turbocharger was originally fitted with 16 mm [0.630 in] capscrews, 16 mm [0.630 in] B16 capscrews with lockplates underneath **must** be used for replacement, available for order as a service kit. On turbochargers that do **not** use the CSeal, a 0.25 mm [0.010 in] shim **must** be fitted between the flange of the bearing housing and the turbine housing. See Figure

3 (ordered separately). Turbochargers that use the CSeal **must not** use a shim between the flange of the bearing housing and turbine housing.

- 25 mm [0.984 in] capscrews of either B16 or Inconel material and 16 mm [0.630 in] capscrews installed on CSeal turbochargers requires double torquing of the bolts which is critical to verifying the proper torque retention of the bolted joint. 16 mm [0.630 in] capscrews installed on turbochargers that do **not** use a CSeal do **not** require double torquing of the bolts. Torque the capscrews in the sequence shown in Figure 4 with the torque values shown in Table 1.
- Turbocharger capscrew replacements can be completed in-chassis if service access is available to complete the torquing sequence appropriately. If access is inhibited, remove turbocharger to make this repair.
- If 25 mm [0.984 in] Inconel capscrews are unavailable, the recommended countermeasure is to install 25 mm [0.984 in] B16 capscrews with lockplates. The same length capscrew **must** be used for replacement. See Table 1.



Figure 2, Lockplate Bent Tab Requirements: Left: Tab Bent Over Capscrew Head Corner (Unacceptable) Right: Tab Bent Over Capscrew Head Flat (Acceptable).

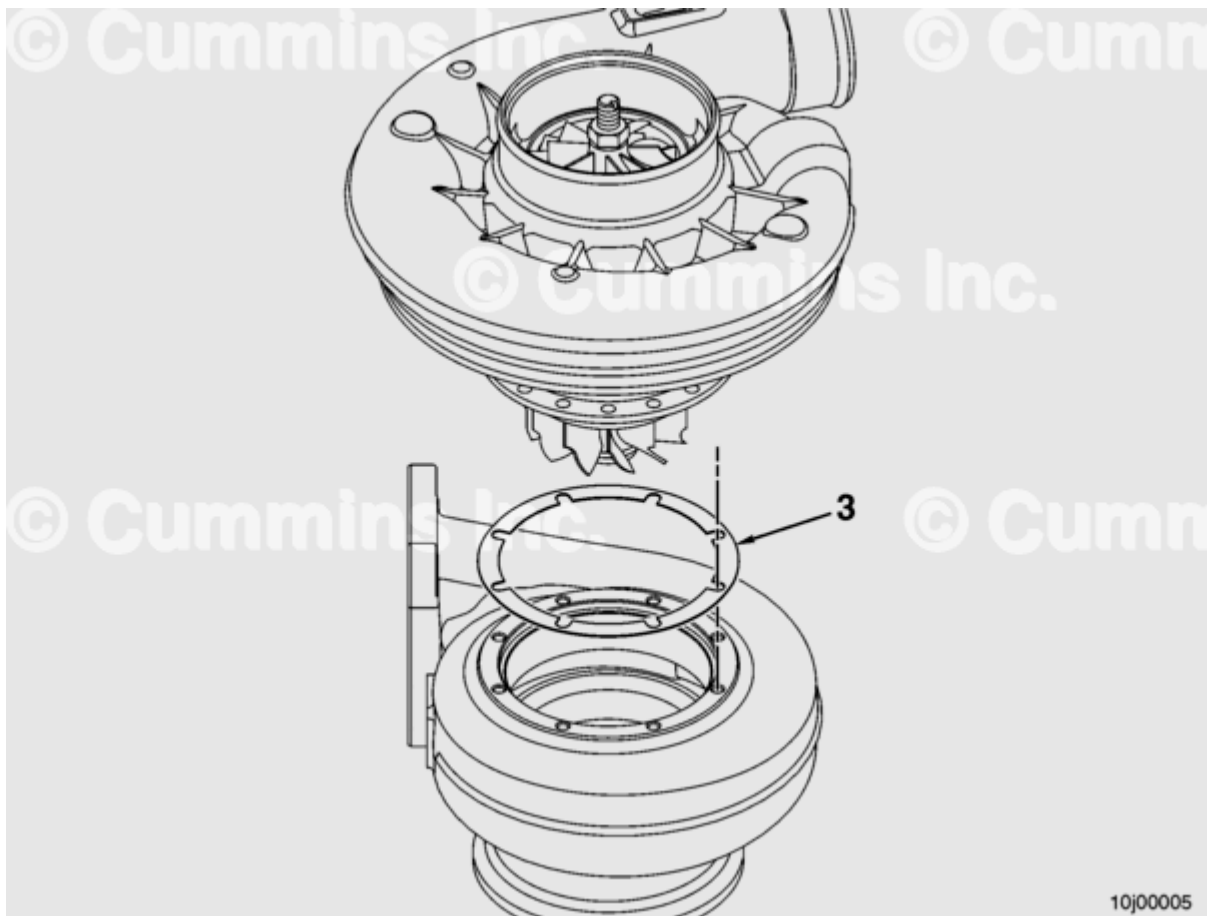


Figure 3, Shim Location

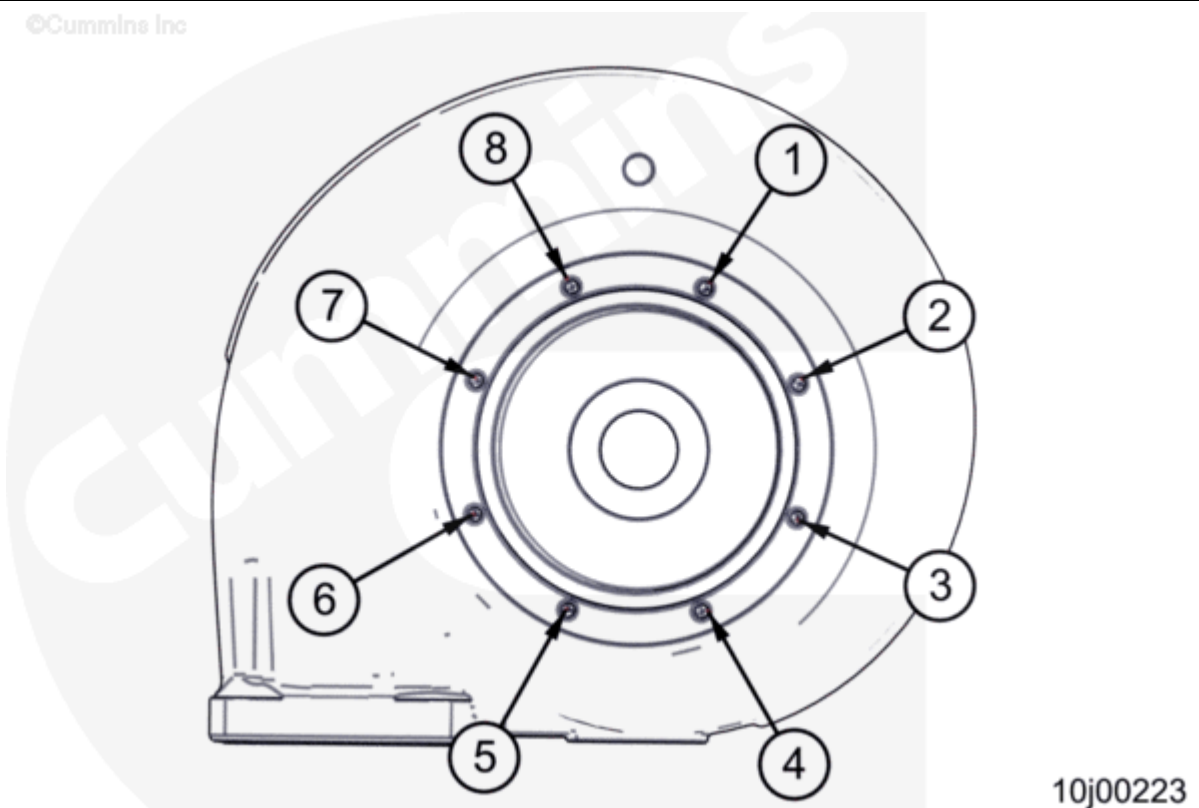


Figure 4, Capscrews Torque Sequence.

Tighten capscrews as shown in Figure 4.

Service Parts

Service parts are offered. See Table 1 for part numbers.

Table 1, Service Parts and Torque Values

Applicable Service Model Names	Bolted Joint Type	Torque Value	Service Part(s)	Service Part Description
QSK60 CM500 QSK60 CM850 MCRS QSK60 CM2150 MCRS	16 mm [0.630 in] B16 with Lockplate	1. 20.3 N•m [15.0 ft-lb] 2. 20.3 N•m [15.0 ft-lb] + Up to 60° additional	5357484	Kit, Service (contains 8 Capscrews and 4 Lockplates)
	25 mm [0.984 in] B16 with Lockplate	1. 26 N•m [19.2 ft-lb] 2. 26 N•m [19.2 ft-lb] + Up to 60° additional	5460104	25 mm [0.984 in] B16 Capscrew (Individually Sold - quantity 8 needed)
			3522904	Lockplate (Individually Sold - quantity 4 needed)
	16 mm [0.630 in] B16 with Lockplate and Shim (Non-CSeal Turbocharger Only)	1. 20.3 N•m [15.0 ft-lb] + Up to 60° additional	5357484	Kit, Service (contains 8 Capscrews and 4 Lockplates)
			3780450	Shim (quantity 1 needed)
	25 mm [0.984 in] Inconel	1. 40 N•m [30 ft-lb] 2. 40 N•m [30 ft-lb] + Up to 60° additional	6417369	25 mm [0.984 in] Inconel Capscrew Service Kit (Contains 8 Capscrews)
QSK60 CM2350 K116 QSK60 CM2350 K121 QSK60 CM2350 K135 QSK60 CM2350 K136			3522904	Lockplate (Individually Sold - quantity 4 needed)

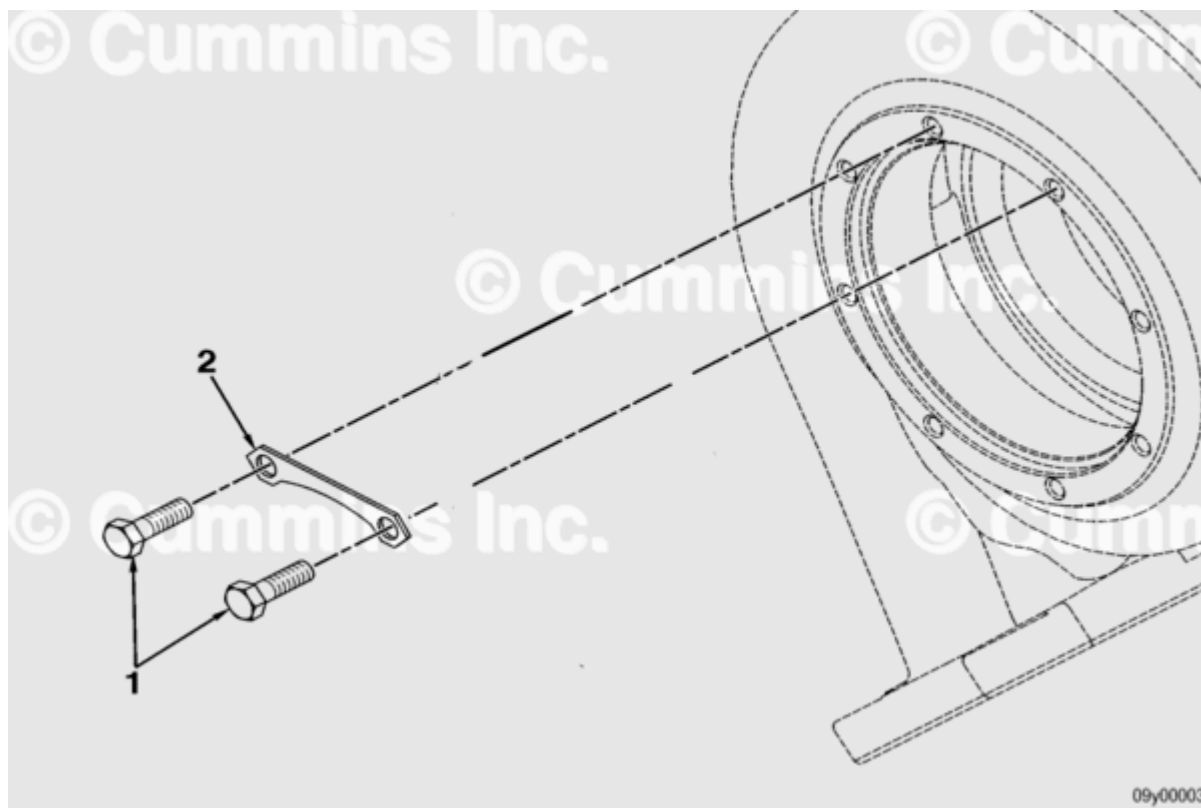


Figure 5, Visual Example of Bolted Joint Architecture for 16 mm [0.630 in] B16 with Lockplate and 25 mm [0.984 in] B16 with Lockplate turbochargers for QSK60 CM500, QSK60 CM850 MCRS, and QSK60 CM2150 MCRS Service Model Names

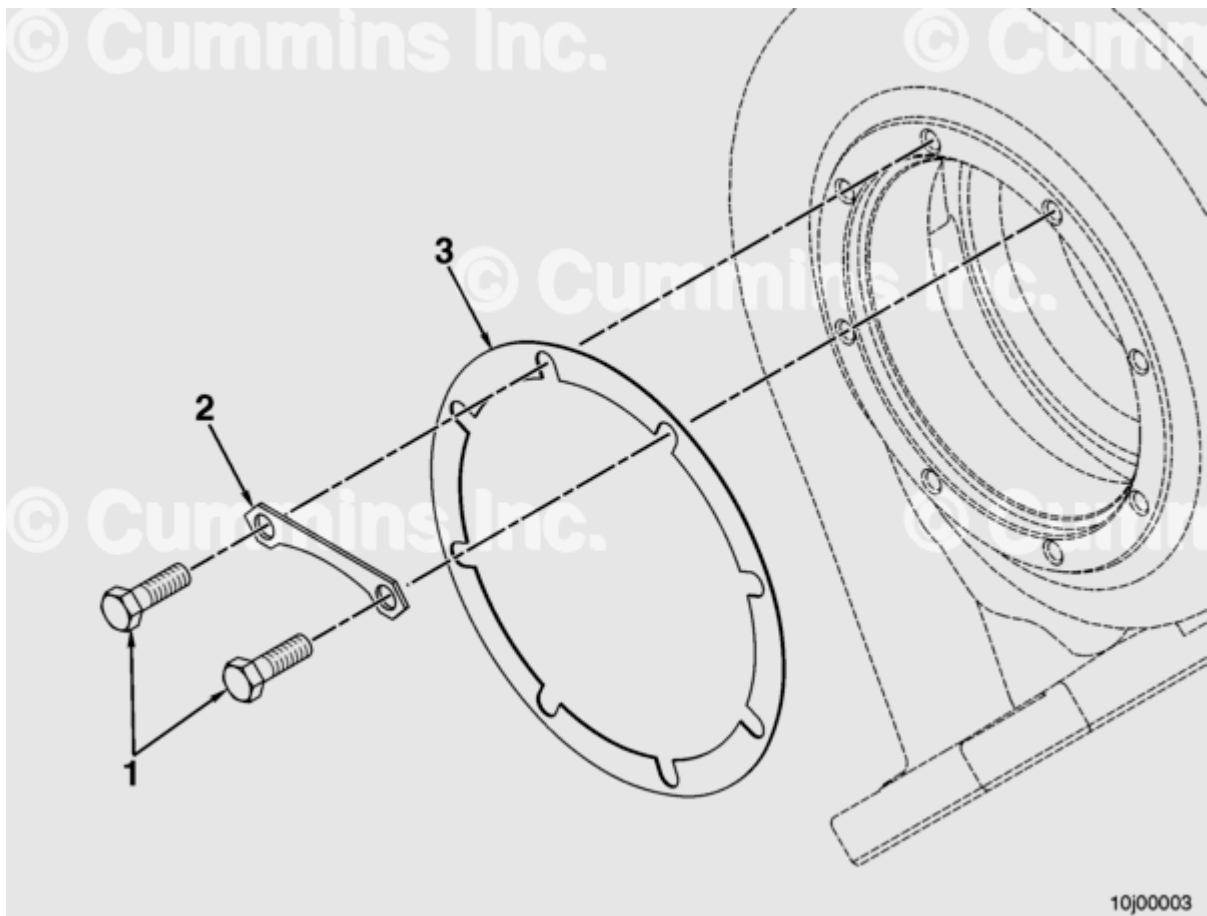


Figure 6, Visual Example of Bolted Joint Architecture for 16 mm [0.630 in] B16 with Lockplate and Shim (Non-CSeal Turbo Only) turbochargers for QSK60 CM500, QSK60 CM850 MCRS, and QSK60 CM2150 MCRS Service Model Names

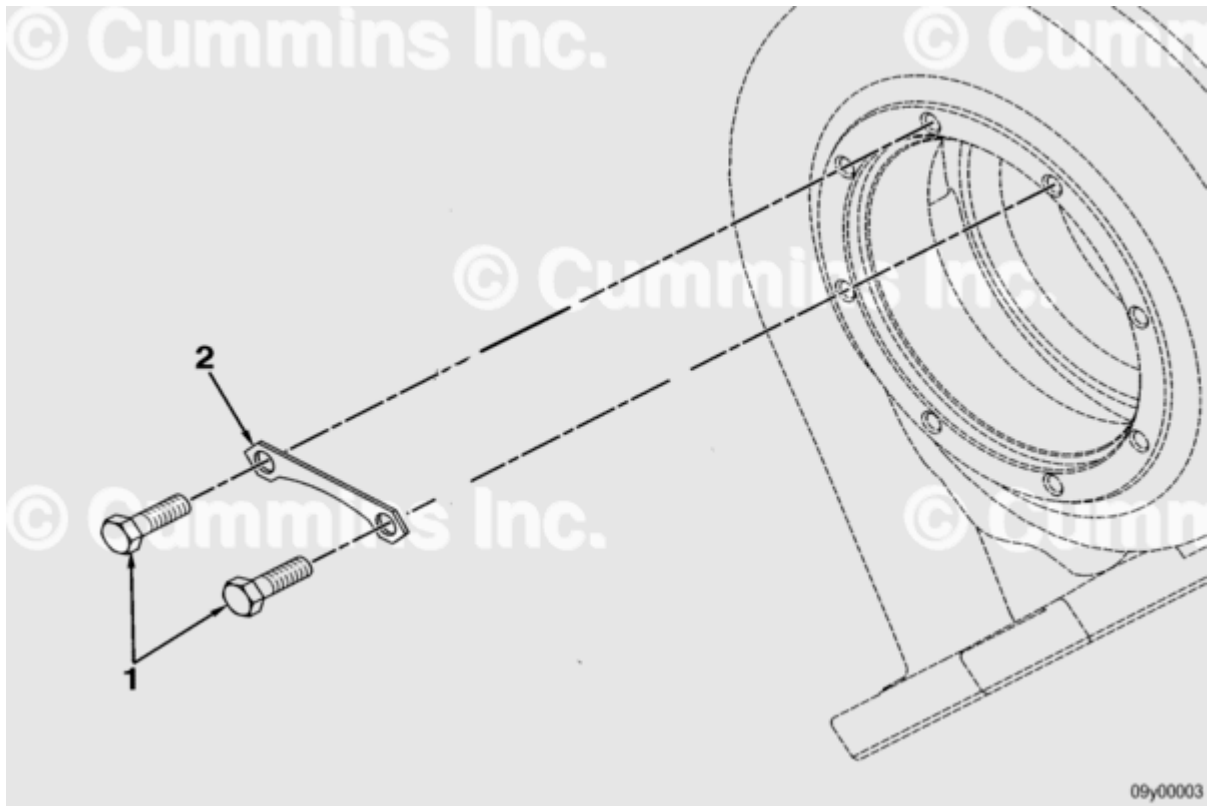


Figure 7, Visual Example of Bolted Joint Architecture for 25 mm [0.984 in] Inconel with Lockplate turbochargers for QSK60 CM2350 K116, QSK60 CM2350 K121, QSK60 CM2350 K135, and QSK60 CM2350 K136 Service Model Name

Document History

Date	Details
2022-11-11	Module Created
2024-6-13	Product Problem Solving (PPS) Step 3 Action
2024-7-2	Updated Table 1.
2026-3-9	Non-Product Problem Solving (PPS)
2026-4-2	Non-Product Problem Solving (PPS)

Last Modified: 15-Apr-2026